

Elizabeth C. Crites, Ph.D.

CONTACT INFORMATION elizabeth_crites@alumni.brown.edu
elizabeth-crites.github.io

CITIZENSHIP Canada, US, UK

CURRENT ROLE  **Parity Technologies** 2026 –
Research Scientist

PAST ROLES  **Web3 Foundation** 2023 – 2025
Research Scientist

 **The University of Edinburgh**, Edinburgh, UK 2021 – 2023
Research Associate

 **Input Output (IOG/IOHK)** 2021
Research Fellow

 **University College London (UCL)**, London, UK 2019 – 2021
Research Fellow

EDUCATION  **Brown University**, Providence, USA
Ph.D. & M.Sc. in Mathematics
Advisor: Anna Lysyanskaya

 **Columbia University in the City of New York**, New York, USA
M.Sc. in Applied Mathematics
Advisors: Richard S. Hamilton & Michael I. Weinstein

 **The University of Western Ontario**, London, Canada
B.Sc. Honours Specialization in Mathematics, with Distinction

 **McGill University**, Montréal, Canada
Visiting Scholar, Honours Mathematics

A full course list can be found [here](#).

AWARDS & SCHOLARSHIPS Best Early Career Paper Award, CRYPTO 2023
US Department of Veterans Affairs Scholarship
Columbia University Admission Scholarship
The University of Western Ontario Admission Scholarship

PUBLICATIONS **On Reed-Solomon Proximity Gaps Conjectures**
Elizabeth Crites, Alistair Stewart
Disproves several conjectures underpinning the security and efficiency of FRI-based SNARKs.
CRYPTO 2026. <https://eprint.iacr.org/2025/2046>

On the Adaptive Security of Key-Unique Threshold Signatures
Michele Ciampi, Elizabeth Crites, Chelsea Komlo, Mary Maller
Impossibility results ruling out adaptive security for large classes of threshold signatures.
CRYPTO 2026. <https://eprint.iacr.org/2025/943>

UC4Free! Existing Threshold Signatures are UC Secure

Jan Bobolz, Elizabeth Crites, Markulf Kohlweiss, Akira Takahashi

A hierarchy of game-based security definitions with proven UC equivalence.

EUROCRYPT 2026. <https://eprint.iacr.org/2026/911>

On the Adaptive Security of FROST

Elizabeth Crites, Jonathan Katz, Chelsea Komlo, Stefano Tessaro, Chenzhi Zhu

Provable security of the FROST threshold signature scheme under adaptive corruptions.

CRYPTO 2025. <https://eprint.iacr.org/2025/1061>

A Plausible Attack on the Adaptive Security of Threshold Schnorr Signatures

Elizabeth Crites, Alistair Stewart

A plausible efficient attack on the adaptive security of a wide range of threshold Schnorr signatures.

CRYPTO 2025. <https://eprint.iacr.org/2025/1001>

Sybil-Resilient Anonymous Signatures with Applications to Decentralized Identity

Elizabeth Crites, Markulf Kohlweiss, Aggelos Kiayias, Amirreza Sarencheh

A VRF-based digital identity solution that is Sybil resilient, anonymous, non-interactive, and stateless.

CCS 2025. <https://eprint.iacr.org/2024/379>

Sassafras: Efficient Batch Single Leader Election

Jeffrey Burdges, Elizabeth Crites, Handan Kılınç Alper, Alistair Stewart, Sergey Vasilyev

An efficient single leader election protocol from ring verifiable random functions.

ACNS 2025. <https://eprint.iacr.org/2023/031>

Ring Verifiable Random Functions and Zero-Knowledge Continuations

Jeffrey Burdges, Oana Ciobotaru, Elizabeth Crites, Handan Kılınç Alper, Alistair Stewart, Sergey Vasilyev

Combining verifiable random functions and ring signatures for leader election and digital identity.

<https://eprint.iacr.org/2023/002>

Fully Adaptive Schnorr Threshold Signatures

Elizabeth Crites, Chelsea Komlo, Mary Maller

First Schnorr threshold signature scheme secure against maximal adaptive corruption.

CRYPTO 2023. Best Early Career Paper Award. <https://eprint.iacr.org/2023/445>

Snowblind: A Threshold Blind Signature in Pairing-Free Groups

Elizabeth Crites, Chelsea Komlo, Mary Maller, Stefano Tessaro, Chenzhi Zhu

First threshold blind signature scheme in pairing-free groups.

CRYPTO 2023. <https://eprint.iacr.org/2023/1228>

Threshold Structure-Preserving Signatures

Elizabeth Crites, Markulf Kohlweiss, Bart Preneel, Mahdi Sedaghat, Daniel Slamanig

First threshold structure-preserving signature scheme.

ASIACRYPT 2023. <https://eprint.iacr.org/2022/839>

Better than Advertised Security for Non-Interactive Threshold Signatures

Mihir Bellare, Elizabeth Crites, Chelsea Komlo, Mary Maller, Stefano Tessaro, Chenzhi Zhu

Security analysis for the FROST and BLS threshold signature schemes.

CRYPTO 2022. Springer link

How to Prove Schnorr Assuming Schnorr: Security of Multi- and Threshold Signatures

Elizabeth Crites, Chelsea Komlo, Mary Maller

Efficient two- and three-round multi- and threshold Schnorr signatures.

Results included in the FROST IRTF RFC 9591. <https://eprint.iacr.org/2021/1375>

Mercurial Signatures for Variable-Length Messages

Elizabeth C. Crites, Anna Lysyanskaya

Extended mercurial signatures to allow messages of unbounded length (e.g., credential attributes).

PETS 2021. <https://eprint.iacr.org/2020/979>

Reputable List Curation from Decentralized Voting

Elizabeth C. Crites, Mary Maller, Sarah Meiklejohn, Rebekah Mercer

Constructed a token-curated registry from a voting protocol with ballot secrecy.

PETS 2020. <https://eprint.iacr.org/2020/709>

Delegatable Anonymous Credentials from Mercurial Signatures

Elizabeth C. Crites, Anna Lysyanskaya

Constructed first efficient scheme for issuing, presenting, and delegating credentials anonymously.

CT-RSA 2019. <https://eprint.iacr.org/2018/923>

DOCTORAL
DISSERTATION

Delegatable Anonymous Credentials from Mercurial Signatures

Introduced a new type of digital signature, called a mercurial signature, and constructed first efficient delegatable anonymous credential (DAC) scheme. Extended mercurial signatures to allow messages of unbounded length. Constructed DAC scheme for multiple certification authorities.

Brown University Library 2019. <https://repository.library.brown.edu/studio/item/bdr:918764/>

MASTER'S
RESEARCH

Conducted research on partial differential equations, such as mean curvature flow and the Ricci flow, used in Richard S. Hamilton's program for solving the Poincaré Conjecture (Millennium Prize Problem). Advisor: Richard S. Hamilton

ACTIVITIES
AND
SERVICE

NIST Call for Multi-Party Threshold Schemes

Team member submitting to the U.S. National Institute of Standards and Technology (NIST) call for multi-party threshold schemes.

Research Workshop on Foundations and Applications of Zero-Knowledge Proofs

Organizer, International Centre for Mathematical Sciences (ICMS), Edinburgh, 2024.

CrossFyre 2024

Organizer, 13th International Workshop on Cryptography, Robustness, and Provably Secure Schemes for Female Young Researchers, EUROCRYPT Affiliated Event, Zurich, 2024.

ZK-Lab

Member, The University of Edinburgh, 2023-2024.

Program Committees

I am or have been a Program Committee member of SCN 2026, the Workshop on Cryptographic Tools for Blockchains (CTB 2026), EUROCRYPT 2025, ASIACRYPT 2024 (*Distinguished PC Members Award*), CRYPTO 2024, EUROCRYPT 2024, the 27th Information Security Conference (ISC 2024), the 1st Workshop on Proofs and Proof Techniques for Cryptographic Security (ProTeCS 2024), and the Institute of Mathematics and its Applications (IMA) International Conference on Cryptography and Coding (IMACC 2023).

I am or have been a reviewer for the following conferences and journals: CRYPTO, EUROCRYPT, Security and Cryptography for Networks (SCN), Designs, Codes and Cryptography (DESI), ACM Transactions on Privacy and Security (TOPS), Applied Cryptography and Network Security (ACNS), IEEE International Conference on Distributed Computing Systems (ICDCS), ACM Advances in Financial Technologies (AFT).

PRESENTATIONS

NIST Workshop on Multi-Party Threshold Schemes 2026

“On the Adaptive Security of Threshold Schnorr Signatures: New Frontiers”

zkStudyClub 2025

“Reed Solomon Proximity Gaps & Disproving Existing Conjectures”

CRYPTO 2025, Santa Barbara, USA

“On the Adaptive Security of FROST”

CRYPTO 2025, Santa Barbara, USA

“A Plausible Attack on the Adaptive Security of Threshold Schnorr Signatures”

Web3 Summit 2024, Berlin, Germany

“Recent Developments on Multi-Party Schnorr Signatures”

CRYPTO 2023, Santa Barbara, USA

“Fully Adaptive Schnorr Threshold Signatures” (Best Paper Plenary)

CrossFyre 2023, Lyon, France

“Multi-Party Schnorr Signatures”

Real World Crypto 2023, Tokyo, Japan

“From Theory to Practice to Theory: Lessons Learned in Multi-Party Schnorr”

London Crypto Day 2022, London, UK

“Recent Developments on Multi-Party Schnorr Signatures”

IOG - UEdinburgh Research Week 2022, Edinburgh, UK

“Multi-Party Schnorr Signatures”

CRYPTO 2022, Santa Barbara, USA

“Better than Advertised Security for Non-Interactive Threshold Signatures”

Zcon3 Conference 2022, Las Vegas, USA

“Research Updates on FROST”

Future of PI: Challenges and Perspectives of Personal Identification 2021

“Delegatable Anonymous Credentials from Mercurial Signatures”

IEEE European Symposium on Security and Privacy (EuroS&P)

University of Waterloo Cryptography, Security, and Privacy Seminar 2021

“Delegatable Anonymous Credentials from Mercurial Signatures”

PETS 2021 Privacy Enhancing Technologies Symposium

“Mercurial Signatures for Variable-Length Messages”

PETS 2020 Privacy Enhancing Technologies Symposium

“Reputable List Curation from Decentralized Voting”

CT-RSA 2019 The Cryptographers' Track at the RSA Conference, San Francisco, USA
“Delegatable Anonymous Credentials from Mercurial Signatures”

TEACHING

COMP0141 Security

Teaching Assistant, University College London

CSCI 1510 Introduction to Cryptography and Computer Security

Teaching Assistant, Brown University

ENGN 1570 Linear System Analysis

Teaching Assistant, Brown University

MATH 0100 Introductory Calculus, Part II

Teaching Assistant, Brown University

MATH 0520 Linear Algebra

Teaching Assistant, Brown University

PAST
ACTIVITIES

CAPS @ Brown : Cryptography Anonymity Privacy Security

Brown University, Providence, USA

Brown-IMPA Watson Brazil Initiative

Hyperbolic Geometry and Minimal Surfaces

Instituto Nacional de Matemática Pura e Aplicada (IMPA), Rio de Janeiro, Brazil

Brown-Kobe Summer School in High Performance Computing

K computer, 3D visualization of peridynamic theory of fracture in solid mechanics.

Kobe University, Kobe, Japan

The Mathematics Scholars Group

The University of Western Ontario, London, Canada